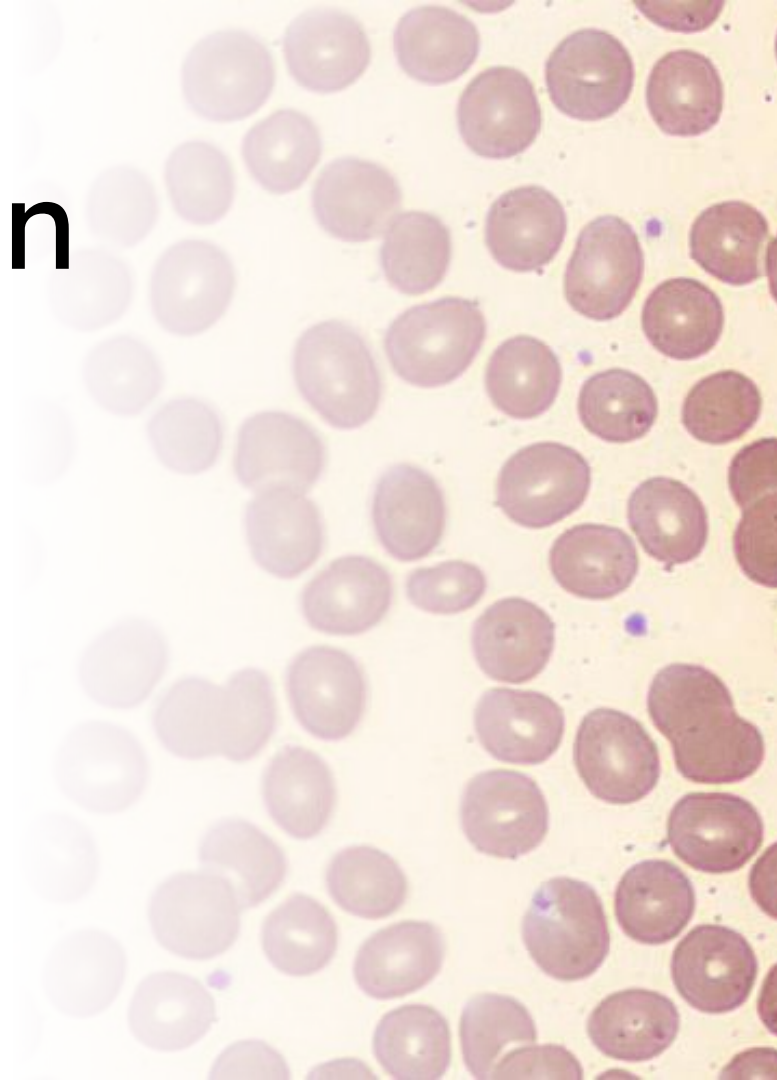


Rouleaux Detection

A Classical Computer Vision Approach

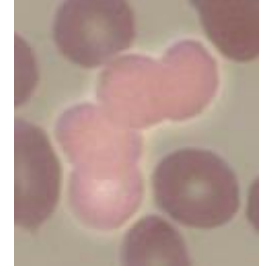
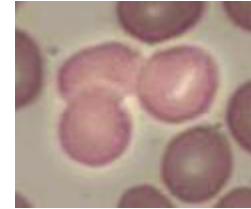
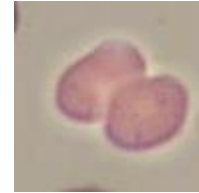
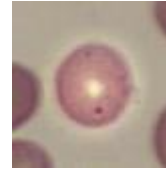
MSKCC Presentation
May 4, 2026

Fernando Aparicio García
MA Statistics 26⁺ - UC Berkeley

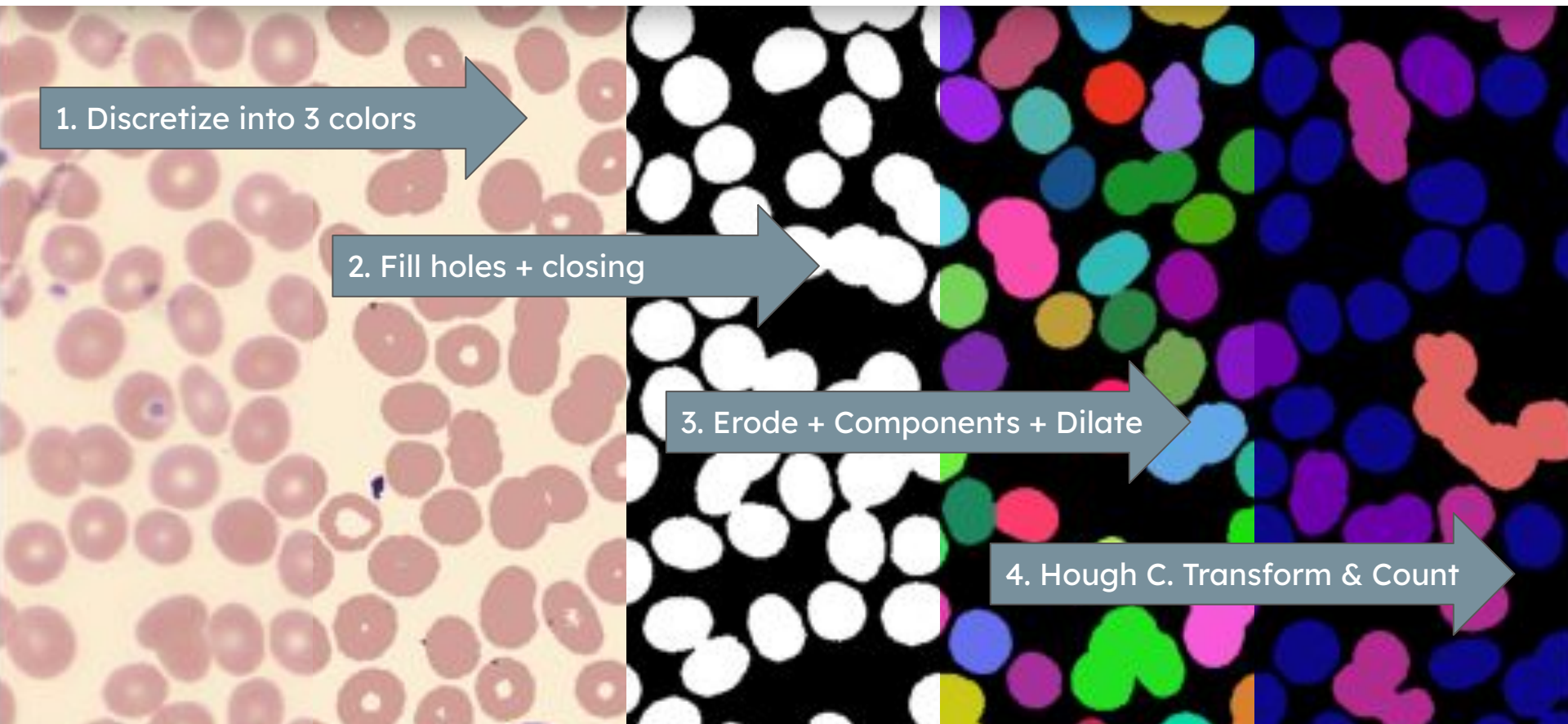


The Goals

- Analyze Rouleaux Formation in Red Blood Cells
 - Discern between single cells and cell clusters
 - On cell clusters, count the number of cells
- Use Classical methods in Computer Vision (no AI/ML)
- Develop a system to label cells manually and use it to elaborate metrics for the method
- Make the pipeline adaptable to other labs



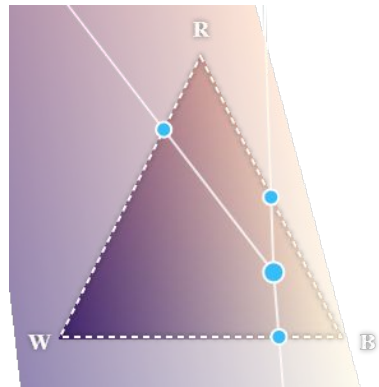
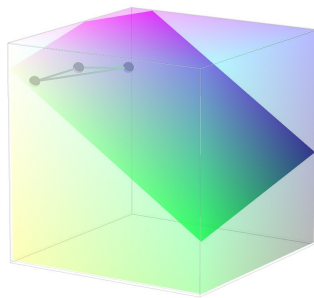
The Pipeline



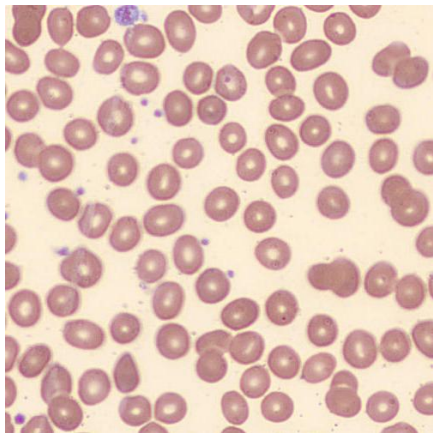
Step 1: Pixel-wise labeling

1.1 Project all colors into a color plane

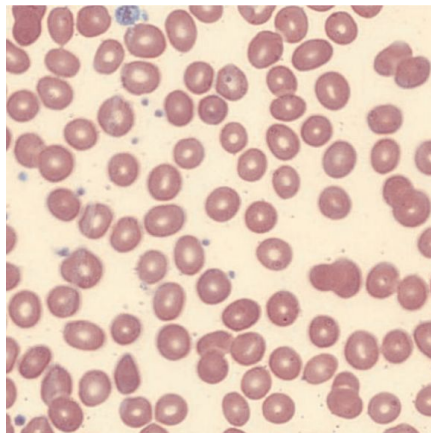
1.2 Perform a voronoi-like separation
into 3 categories



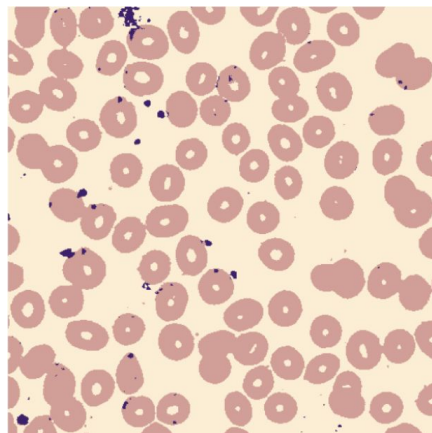
Original Image



Color Projection



Final Discretization



Step 1: Pixel-wise labeling

Pixelwise label tuning

Color Config

UPLOAD IMAGE

Choose File 1_37550251_1500_2000.jpg

RBC

BG

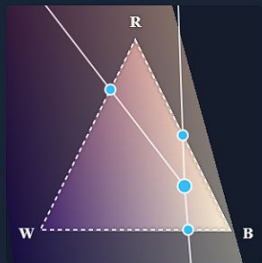
WBC



VIEW MODE

Mask-Plane

Color Triangle



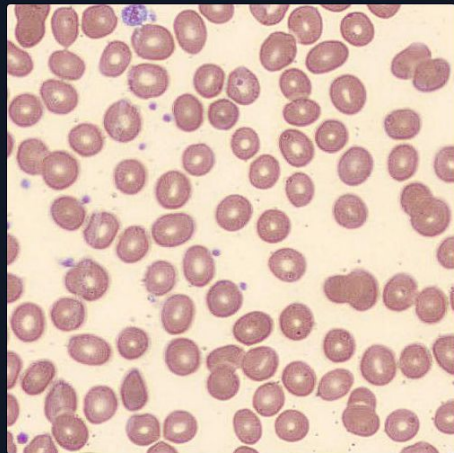
$t_R = 0.78$ $t_B = 0.26$ $t_G = 0.50$
 $v_G = [0.23, 0.65, 0.12]$

Copy JSON

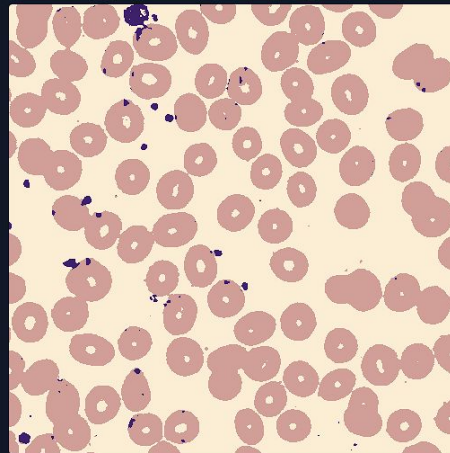
Paste config JSON here...

Load Config

ORIGINAL



PROCESSED



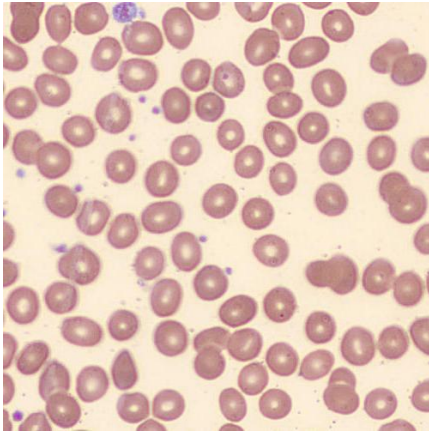
Step 2: Final RBC mask

2.1 Fill holes within Red Blood Cells

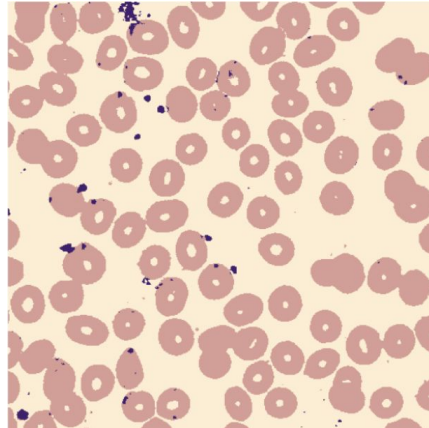
2.2 Opening (erosion+dilation)

- Smooths edges
- Removes salt & pepper (platelets)

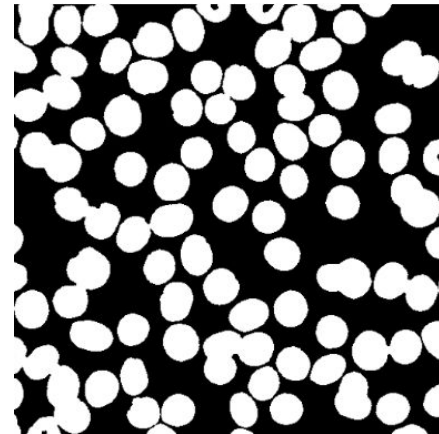
Original Image



Step 1 Output



Final RBC Mask



Step 3: Separating clusters

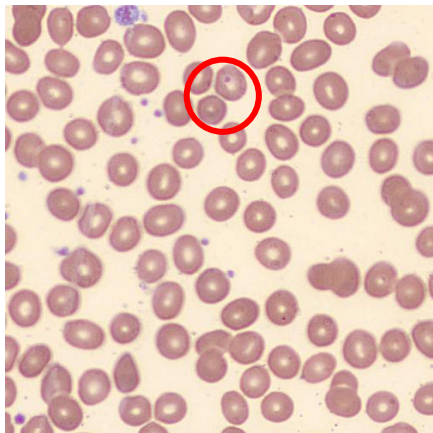
Naive approach: take connected components directly

Inaccurate: Over-clusters!

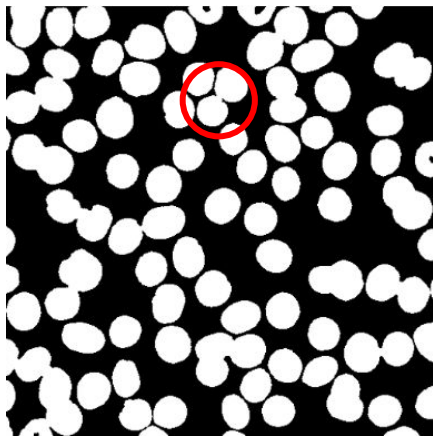
Solution: “conjugate” with an erosion

- Erode, then take connected components, then dilate back

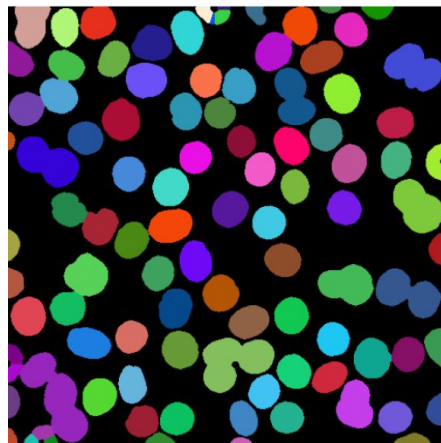
Original Image



Step 2 Output



Final Clusters



Step 4: Identifying/counting Rouleaux

For each separate cluster, perform a Hough Circle transform, then return the number of centers

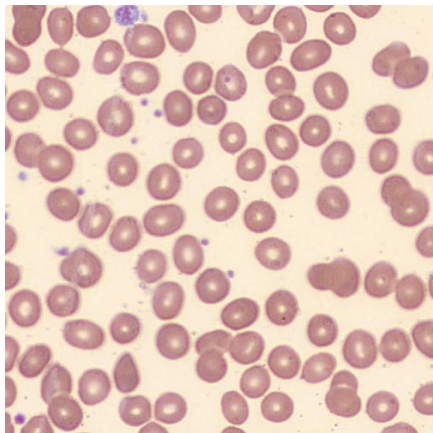
4.1 Canny edge detection

4.2 For each pixel and each radius $r_{\min} < r < r_{\max}$:
Take the %-age of the circle that overlaps with the edges

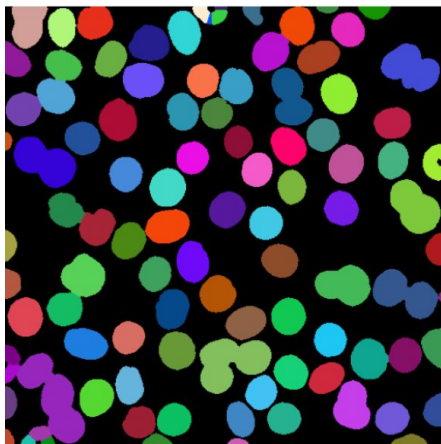
4.3 Discard circles that do not meet a threshold

4.4 Among final circles, return local maxima

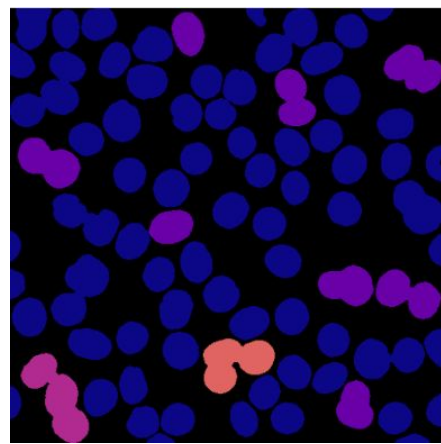
Original Image



Step 3 output



Final Labels

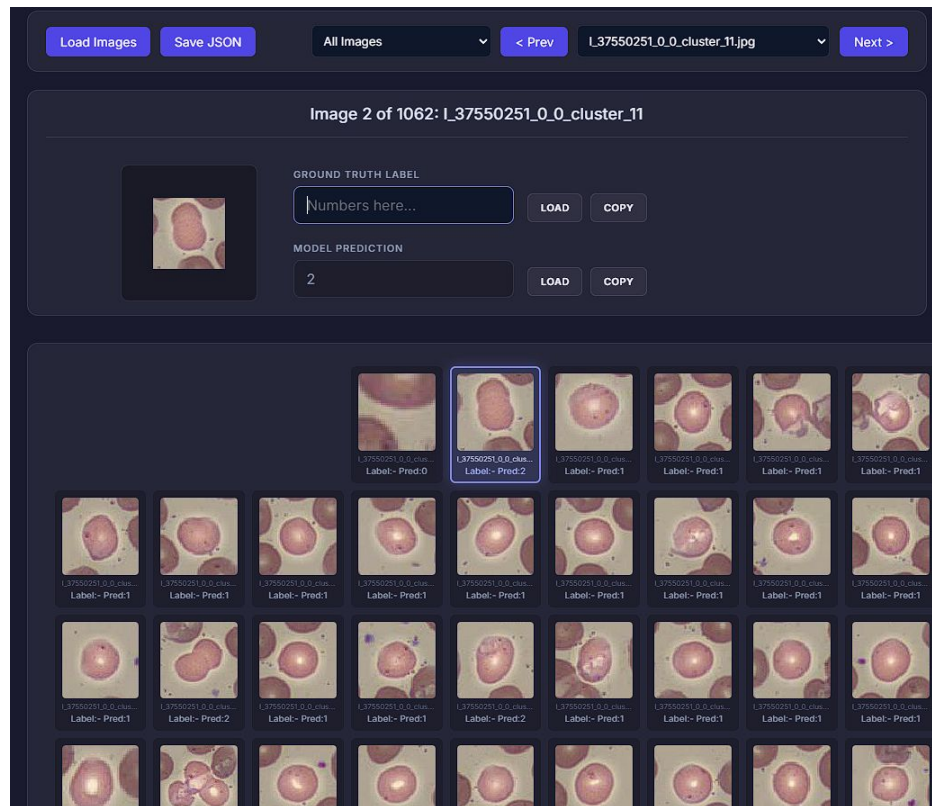


Evaluation

Ground truth: manual labels

Evaluation results:

- Evaluated on 6 test 500x500 slices
- Cluster sensitivity: 90.41%
- Cluster precision: 92.96%
- Cluster specificity: 98.60%
- Cluster errors:
 - MAE 0.05, MSE: 0.05
- Only Rouleaux:
 - MAE 0.16, MSE: 0.16



Thank you!

Any Questions?

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